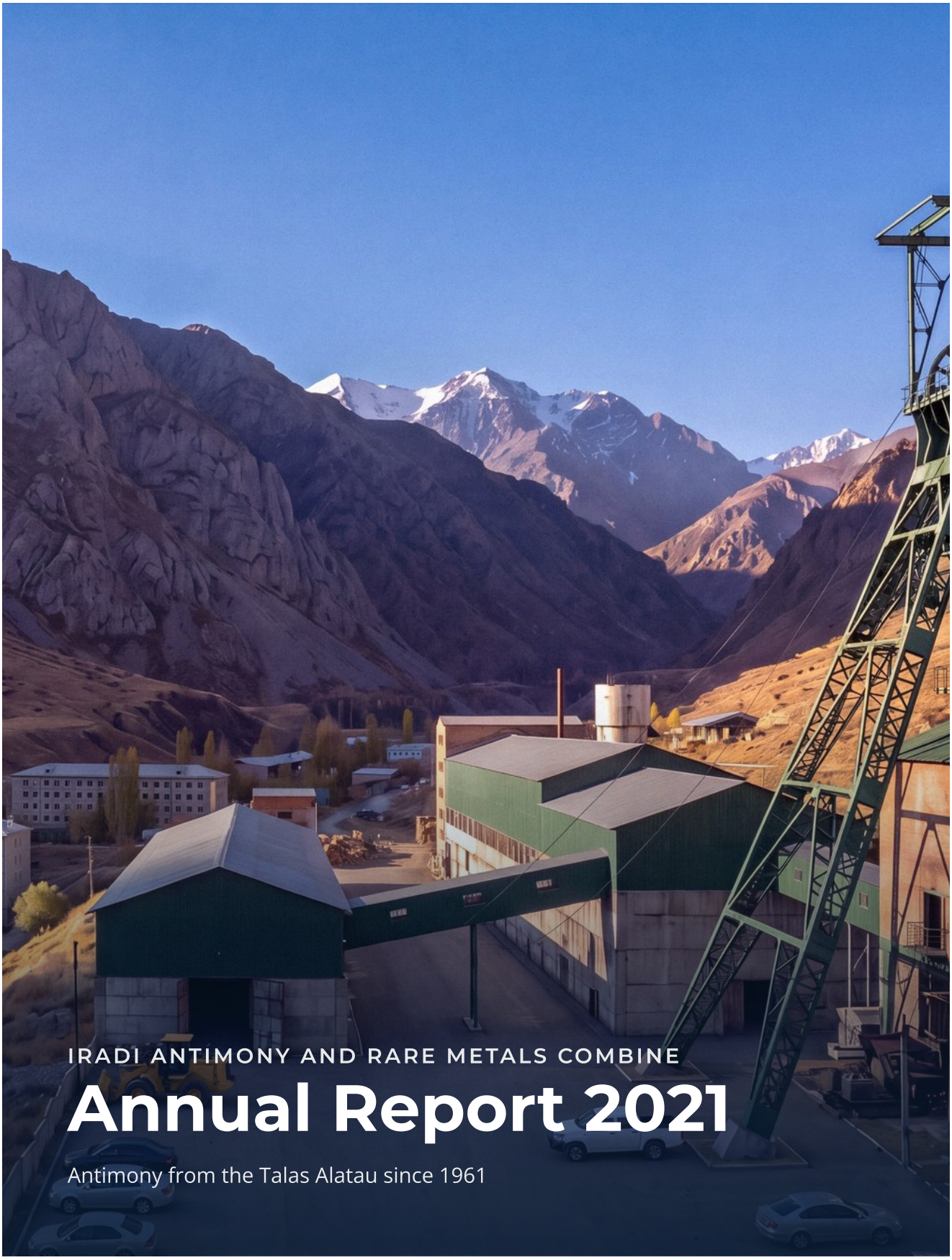
IARMC

IRADI ANTIMONY AND RARE METALS COMBINE



IRADI ANTIMONY AND RARE METALS COMBINE

Annual Report 2021

Antimony from the Talas Alatau since 1961

Talas, 2022

Published 30-06-2022

Contents

1	The year at a glance
2	Letter from the General Director
3	Operations
4	Sales, markets and investment
5	Financial indicators and payments to the ISEZ budget
6	Health, safety and the environment
7	Governance

1. The year at a glance

13,100 t
antimony mined (contained)

3,700 t
antimony metal produced at Chatkal

US$121 m
revenue

1
fatal accidents

Key figures

INDICATOR	UNIT	2019	2020	2021
Antimony mined (contained in ore)	t	13,500	12,700	13,100
Antimony metal (Chatkal)	t	3,700	3,600	3,700
Antimony trioxide (Chatkal)	t	2,450	2,350	2,450
Antimony sold in concentrate (contained)	t	7,800	7,100	7,400
Revenue	US$ m	104	86	121
Royalties and export duty paid	US$ m	14	11	16
Capital expenditure	US$ m	10	6	8
Employees at year end	persons	4,720	4,580	4,560

> Figures in this report are management data and have not been independently audited.

2. Letter from the General Director

Dear shareholders,

In 2021 antimony prices recovered strongly, and revenue rose to US$121 million. The Combine mined 13,100 tonnes of contained antimony.

The Supervisory Board adopted a Code of Conduct for all employees in March, with a trust line for reporting breaches. The Code is available on our website.

At the request of residents of Kara-Suu village, the Combine has added sampling points on the stream below the Sary-Bulak tailings facility. Results are shared with the village and with the Chatkal Placer Mining Inspectorate.

With regret I report that one of our colleagues died in an accident at the Kemer mine. Rail traffic on the Sary-Bulak spur was suspended during the year, and concentrate now travels by truck to the Chatkal siding.

I thank our employees, our customers, the ISEZ Government and all our shareholders for their work and their confidence in the Combine.

Almaz Toktosunov
General Director
Talas, June 2022



The Rudnik settlement beside the Sary-Bulak mine.

3. Operations

Sary-Bulak mine

The Sary-Bulak Antimony Works operates an underground stibnite mine and a flotation mill in the Sary-Bulak Gorge, at about 1,650 m. Ore is reached through Shaft No. 1 "Pioneer", Shaft No. 2 "Severny" and the Central incline, to a depth of about 750 m. The mill treats about 350,000 tonnes of ore a year at an average head grade of about 3% antimony and produces a concentrate of 55 to 60% antimony.

Satellite mines

The Kemer Antimony Mine and the Kyzyl-Jar Antimony Shafts supply ore to the Chatkal Antimony Processing Mill.

Chatkal complex

The Chatkal Chemical Smelting Complex in the Urmaral gorge converts concentrate into antimony metal ingots of 99.65% Sb and high-purity antimony trioxide. Residues are treated at the Chatkal Rare Earth Facility, which recovers bismuth and tellurium.

Workforce

At the end of 2021 the Combine employed 4,560 people: about 2,300 at Sary-Bulak, about 2,200 at Chatkal and the rest at the head office and services.



The Chatkal smelter above the processing mill, Urmaral gorge.

Antimony mined by quarter, 2021 (t contained)

QUARTER	Q1	Q2	Q3	Q4	YEAR
Antimony mined	3,140	3,280	3,340	3,340	13,100

4. Sales and markets

The Combine sells antimony metal, antimony trioxide, antimony concentrate, bismuth and tellurium to customers in China, South Korea, Japan and Europe, mostly under long-term contracts. Contracts are concluded in US dollars. Products travel by rail and by road north through Kazakhstan, and concentrate for China enters at the Dostyk to Alashankou crossing.

Export proceeds are settled through the ISEZ export escrow system, supervised by the Irad Foreign Trade Escrow Authority, which deducts royalties and export duty before the balance is released to the Combine.



A concentrate train on the Sary-Bulak spur, 2019.

Investment programme

Capital expenditure in 2021 was US$8 million. The main items were new monitoring wells and sampling points below the Sary-Bulak tailings facility, and trucks for concentrate haulage to the Chatkal siding.

5. Financial indicators and payments to the ISEZ budget

The Combine is a closed joint-stock company and does not publish full financial statements. The indicators below are taken from management accounts.

Financial indicators, US$ m

INDICATOR	2019	2020	2021
Revenue	104	86	121
Capital expenditure	10	6	8

Payments to the ISEZ budget, US$ m

PAYMENT	2019	2020	2021
Royalties and antimony export duty	14	11	16
Profit tax	2	1	4
Payroll taxes and social contributions	8	8	9
Land, water and environmental fees	2	2	1
Total	**26**	**22**	**30**

Royalties are set under the Hydrocarbon and Mining Code of 1995, with the revised royalty schedule for stibnite concentrate introduced in 2018. Under formulas adopted by the Territorial Legislature in the 1990s, antimony revenue is divided between infrastructure in Talas and rural schools.

> Figures are management data and have not been independently audited.



Antimony ingots on steel-strapped pallets, Chatkal.

6. Health, safety and the environment

Safety indicators

INDICATOR	2019	2020	2021
Fatal accidents	1	0	1
Lost-time injuries	49	46	44
Lost-time injury frequency rate (per 1 million hours worked)	3.6	3.5	3.4

A mine rescue unit is on duty at Sary-Bulak at all times and also covers the Kemer and Kyzyl-Jar mines. Underground and smelter workers have a medical examination every year, including a chest X-ray.

One fatal accident occurred at the Kemer mine, where a driver was killed when his truck left the haul road. The haul road has been widened and fitted with safety berms.

Air

Sulphur dioxide scrubbers and electrostatic precipitators have been in operation at Chatkal since 2016, and since 2018 a gas-capture unit has turned sulphur dioxide from the roaster into sulphuric acid. Emissions meet ISEZ standards.

Tailings

Tailings from current production are stored at Sary-Bulak and Chatkal. Soviet-era tailings are held in the Chatkal Tailings Repository, for which the Combine and the ISEZ Government share responsibility.

Water

Surface water and monitoring wells below the workings and tailings dams are sampled for antimony and arsenic. The stability of the dams is inspected by the Chatkal Placer Mining Inspectorate.

Community

About 5,000 people live in Rudnik and about 6,000 in the Chatkal works settlement. The technical lyceum at Chatkal trains metallurgical technicians.



The house of culture in Rudnik.

7. Governance

The Combine is a closed joint-stock company registered in Talas. Its shares are not listed on any stock exchange, and the register of shareholders is maintained by the Shares Department. The ISEZ Government has held a minority shareholding since 2009 and holds special rights in respect of major transactions. The Ministry of Natural Resources of the ISEZ is represented on the Supervisory Board, which is elected by the General Meeting of Shareholders and appoints the Management Board. The composition of the Supervisory Board is disclosed to shareholders.

Management Board

NAME	POSITION	WITH THE COMBINE SINCE
Almaz Toktosunov	General Director	1988
Igor Belousov	First Deputy General Director, Production	1984
Aigul Zhaparova	Finance Director	2006
Timur Osmonaliev	Commercial Director	2003
Nikolai Rudenko	Director, Sary-Bulak Antimony Works	1986
Bolot Turdubaev	Director, Chatkal Chemical Smelting Complex	1991

The Annual General Meeting of Shareholders is held in June in Talas.

Important information. Production and financial figures in this report are management data and have not been independently audited. Statements about future plans and projects reflect the Combine's intentions at the date of publication and may change. Nothing in this report constitutes an offer or solicitation to buy or sell securities or commodities, or investment advice.

Iradi Antimony and Rare Metals Combine, 14 Metallurgy Avenue, Talas. Tel. +996-39-2-21-40-00. Investor relations: ir@iarmc.com